The Empirical Economics of Cooling

A Thermohydraulic and Economic Performance Analysis of Commercial Coolants

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Abstract

Cooling is an increasingly important economic service whose productive contribution depends not merely on installed cooling capacity but on the effective delivery of thermal services under technological, physical, and resource constraints. The economic evaluation of cooling therefore requires an analytical connection between thermophysical properties, thermohydraulic performance, operating costs, and the value of the cooling service ultimately delivered.

This treatise develops an empirical and computational framework for comparing commercial coolants and examining their potential contribution to economic cooling performance. The empirical foundation consists of a manufacturer-reported property dataset containing 137 observations, a product catalogue containing 10 records, and a source register containing 13 sources. The principal comparative cohort comprises five Dynalene HC grades: HC-10, HC-20, HC-30, HC-40, and HC-50.

A reproducible thermohydraulic model is constructed over 22 exact common temperatures and three mass-flow setpoints, producing 330 grade-temperature-flow operating points. The model calculates Reynolds and Prandtl numbers, Darcy friction factors, pressure losses, estimated pumping requirements, heat-capacity rates, and internal convective heat-transfer coefficients where the selected correlation is applicable. No property interpolation or extrapolation is used in constructing the common-temperature performance surface.

The results demonstrate that thermohydraulic performance is multidimensional. At a common temperature and mass-flow rate, the grade with the highest estimated internal heat-transfer coefficient need not have the lowest pressure loss or the highest heat-capacity rate. Performance comparisons are consequently conditional on the operating domain, system geometry, flow constraint, and selected engineering objective.

The treatise subsequently develops a lifecycle-cost framework and connects thermohydraulic performance to the Cooling Dividend Function. However, the available dataset does not contain independent laboratory measurements, application-level economic outcomes, electricity prices, equipment capital costs, or measured cooling-service delivery. Accordingly, the economic Cooling Dividend is formulated as a testable empirical object rather than reported as an estimated monetary result.

The principal conclusion is that the economic value of a coolant cannot be inferred from an isolated thermophysical property. It must be evaluated through the complete chain connecting material properties, delivered cooling, auxiliary resource consumption, lifecycle costs, and the economic value of the resulting service.

The treatise ends with “The End”

Keywords: Economic Cooling; Cooling Dividend Function; Thermohydraulics; Commercial Coolants; Heat Transfer; Pumping Power; Lifecycle Cost; Cooling Architecture; Thermophysical Properties.

Contents

1 Introduction **3**

2 Theoretical Foundation **4**

2.1 Cooling as an Economic Service 4

2.2 The Gross Cooling Dividend 4

2.3 The Net Cooling Dividend 5

3 Data and Evidence Architecture **5**
3.1 Repository and Dataset . . . 5
3.2 Evidence Classification . . . 5
3.3 The Dynalene HC Cohort . . . 6
3.4 Thermophysical Variables . . . 6

4 Descriptive Analysis of Thermophysical Properties **6**
4.1 The 20 Degree Celsius Comparison . . . 6
4.2 Heat-Capacity Rate . . . 7
4.3 The Importance of Temperature Dependence . . . 7

5 Thermohydraulic Model **7**
5.1 Operating Domain . . . 7
5.2 Continuity and Velocity . . . 8
5.3 Reynolds Number . . . 8
5.4 Prandtl Number . . . 8
5.5 Darcy Friction Factor . . . 9
5.6 Pressure Loss . . . 9
5.7 Pumping Power . . . 9
5.8 Heat-Capacity Rate . . . 9
5.9 Convective Heat Transfer . . . 10

6 The Thermohydraulic Performance Surface **10**
6.1 Vector-Valued Performance . . . 10
6.2 Performance at 20 Degrees Celsius . . . 10
6.3 Performance-Surface Ranges . . . 11
6.4 Interpretation of the Surface . . . 11

7 Economic Interpretation of Thermohydraulic Performance **11**
7.1 From Physical Performance to Economic Cost . . . 11
7.2 Lifecycle Cost . . . 11
7.3 Levelised Cost of Cooling . . . 12
7.4 The Economic Decision Rule . . . 12

8 The Cooling Dividend Function: Empirical Connection **13**
8.1 Gross Cooling Dividend . . . 13
8.2 Marginal Cooling Dividend . . . 13
8.3 Net Cooling Dividend . . . 13
8.4 A Coolant-Specific Representation . . . 13

9 Empirical Hypotheses **13**
9.1 Hypothesis 1: Temperature-Dependent Thermophysical Performance . . . 14
9.2 Hypothesis 2: Grade Heterogeneity . . . 14
9.3 Hypothesis 3: Hydraulic-Thermal Trade-Off . . . 14
9.4 Hypothesis 4: Operating-Condition Dependence . . . 14
9.5 Hypothesis 5: Economic Objective Dependence . . . 14

10 Statistical and Computational Validation **14**
10.1 Property-Function Estimation . . . 14
10.2 Validation Metrics . . . 15
10.3 Avoiding Data Leakage . . . 15
10.4 Uncertainty . . . 15

11 Limitations **15**
11.1 Manufacturer-Reported Data . . . 15
11.2 Unspecified Measurement Conditions . . . 16
11.3 Grade Concentration . . . 16
11.4 Simplified Pipe Geometry . . . 16
11.5 Fixed Pump Efficiency . . . 16

11.6 Heat-Transfer Correlation . . . 16
11.7 Constant Properties Within Operating Points . . . 16
11.8 Absence of Economic Outcome Data . . . 16
11.9 No Safety or Compatibility Certification . . . 17

12 Proposed Experimental Validation **17**
12.1 Test Configuration . . . 17
12.2 Operating Conditions . . . 17
12.3 Repeated Measurements . . . 17
12.4 Uncertainty Propagation . . . 17

13 An Expanded Economic Research Architecture **18**
13.1 Cooling Demand . . . 18
13.2 Cooling Architecture . . . 18
13.3 Delivered Cooling . . . 18
13.4 Economic Outcomes . . . 18
13.5 Future Megacity Panel . . . 18

14 General Findings and Propositions **19**
14.1 Finding 1: Thermophysical Properties Are Multidimensional . . . 19
14.2 Finding 2: Performance Is Operating-Condition Dependent . . . 19
14.3 Proposition 1: Conditional Thermohydraulic Ranking . . . 19
14.4 Proposition 2: The Economic Ranking Is Objective-Dependent . . . 19
14.5 Proposition 3: Thermohydraulic Equivalence Is Conditional . . . 20
14.6 Proposition 4: Economic Cooling Requires Service-Normalized Evaluation . . . 20
14.7 Proposition 5: The Cooling Dividend Is Not a Thermophysical Property . . . 20

15 Conclusion **20**

List of Tables

1 Empirical and Computational Dataset . . . 5
2 Dynalene HC Thermophysical Properties at 20°C . . . 6
3 Model-Estimated Performance at 20°C and 0.3 kg/s . . . 10

1 Introduction

Cooling is a productive economic service. It supports industrial production, commercial activity, residential welfare, digital infrastructure, transportation, energy systems, and the preservation of physical capital.

Its economic significance cannot be evaluated exclusively through the nominal capacity of a cooling installation. The relevant economic question is whether a cooling architecture delivers the required thermal service at an acceptable resource cost and with sufficient reliability.

The preceding theoretical development established the concept of economic cooling and introduced the Cooling Dividend Function as a representation of the incremental economic value attributable to cooling.

The present treatise develops an empirical and computational foundation for investigating the engineering and economic dimensions of that framework.

The investigation begins with commercial coolant properties. A coolant's specific heat capacity, thermal conductivity, density, and viscosity influence the thermal and hydraulic behaviour of a cooling system. These properties jointly determine heat-capacity rate, flow characteristics, pressure loss, pumping requirements, and convective heat-transfer performance.

However, no individual property provides a complete measure of economic usefulness.

A fluid with relatively high thermal conductivity may also impose different hydraulic requirements. A fluid with a high specific heat capacity may provide a greater heat-capacity rate at a given mass flow, but its system-level performance remains conditional on the heat exchanger, operating temperature, pump characteristics, and application.

The relevant comparison is therefore not simply between fluids. It is between fluid-system configurations delivering a specified cooling service under declared operating conditions.

This treatise has five principal objectives:

1. To construct a source-audited empirical foundation from manufacturer-reported thermophysical properties.
2. To compare the temperature-dependent properties of five commercial Dynalene HC grades.
3. To construct a reproducible thermohydraulic performance surface over common temperatures and mass-flow conditions.
4. To formulate an economic lifecycle-cost framework for comparing coolant-system configurations.
5. To establish the empirical connection between thermohydraulic performance and the Cooling Dividend Function.

The distinction between observed properties and model-estimated engineering outcomes is maintained throughout the treatise.

The treatise does not claim to report independent laboratory validation. Nor does it claim to estimate the monetary Cooling Dividend from the available engineering dataset.

2 Theoretical Foundation

2.1 Cooling as an Economic Service

Let an economy employ a cooling service represented by:

$$C \geq 0. \tag{1}$$

The delivered service depends on cooling capital, energy, working fluids, water, operating technology, and system architecture.

A general representation is:

$$C = \Phi(K_C, E_C, W_C, R_C, \mathcal{A}), \tag{2}$$

where:

$$K_C : \text{ cooling capital,} \tag{3}$$
$$E_C : \text{ energy allocated to cooling,} \tag{4}$$
$$W_C : \text{ water and associated resources,} \tag{5}$$
$$R_C : \text{ other operating resources,} \tag{6}$$
$$\mathcal{A} : \text{ cooling architecture.} \tag{7}$$

The delivered cooling service is distinct from installed cooling capacity. An installation may possess substantial nominal capacity while delivering a smaller effective service because of hydraulic constraints, thermal losses, equipment limitations, or operational inefficiencies.

2.2 The Gross Cooling Dividend

Let $Y(C)$ denote the economic output or value generated under a specified cooling configuration.

The Gross Cooling Dividend is defined as:

$$D_C(C) = Y(C) - Y(0). \tag{8}$$

The counterfactual $Y(0)$ must be defined with respect to a specified system boundary, time horizon, and set of controlled conditions.

The expression does not imply that all economic outcomes can be attributed to cooling without an identification strategy.

2.3 The Net Cooling Dividend

Let $SC(C)$ represent the incremental social opportunity cost of providing cooling.

The Net Cooling Dividend is:

$$N_C(C) = D_C(C) - SC(C). \tag{9}$$

The opportunity-cost measure must be constructed carefully. External costs already incorporated in resource prices or other components of the valuation should not be counted again.

The empirical analysis of commercial coolants contributes to this framework by investigating one part of the production technology underlying delivered cooling.

3 Data and Evidence Architecture

3.1 Repository and Dataset

The empirical materials are maintained in the EconomicCooling research repository:

https://github.com/TheRealoliGARCH/EconomicCooling

The repository separates product identification, thermophysical observations, source provenance, and model-generated performance outputs.

The principal data objects are summarized in Table 1.

Table 1: Empirical and Computational Dataset

Data object	**Records**
Master coolant observations	137
Product catalogue	10
Source register	13
HC performance-surface operating points	330
Common exact temperatures	22
Mass-flow setpoints	3
Dynalene HC grades	5

The 137 observations are not 137 independent laboratory experiments. They are extracted manufacturer-reported property observations associated with particular commercial formulations and thermodynamic states.

The product catalogue includes exact formulation records and broader product-family discovery records. Family-level records are not treated as exact quantitative formulations.

The master dataset includes observations associated with DOWFROST HD, Therminol 75, and the five Dynalene HC grade series. The expanded Dynalene HC cohort is the principal focus of the present comparative analysis.

3.2 Evidence Classification

The evidence architecture distinguishes among:

Manufacturer-reported properties: Values transcribed from manufacturer technical documents.

Model-estimated outputs: Quantities calculated using specified engineering equations and assumptions.

Independent measurements: Laboratory measurements conducted independently of the manufacturer.

Experimental validation: Controlled tests designed to evaluate the accuracy of the model.

The present study uses the first two categories. It does not contain independent laboratory measurements or completed experimental validation.

This distinction is fundamental to the interpretation of the results.

3.3 The Dynalene HC Cohort

The comparative cohort consists of:

1. Dynalene HC-10;
2. Dynalene HC-20;
3. Dynalene HC-30;
4. Dynalene HC-40;
5. Dynalene HC-50.

The HC-10, HC-20, HC-30, and HC-40 property observations are sourced from the 2025 Dynalene HC Engineering Guide.

The HC-50 observations in the current master dataset are sourced from the 2020 Dynalene HC Technical Data Sheet.

The source revisions are therefore not harmonized.

The grade suffixes are treated as commercial identifiers. Their numerical values are not interpreted as disclosed mass fractions or volume fractions because the relevant source tables do not establish such a concentration basis.

3.4 Thermophysical Variables

The principal thermophysical variables are:

$$c_{p,g}(T) : \text{specific heat capacity}, \tag{10}$$
$$k_g(T) : \text{thermal conductivity}, \tag{11}$$
$$\rho_g(T) : \text{density}, \tag{12}$$
$$\mu_g(T) : \text{dynamic viscosity}, \tag{13}$$

where g denotes the coolant grade and T denotes temperature.

The corresponding units are:

$$c_p : \mathrm{kJ\,kg^{-1}\,K^{-1}}, \tag{14}$$
$$k : \mathrm{W\,m^{-1}\,K^{-1}}, \tag{15}$$
$$\rho : \mathrm{kg\,m^{-3}}, \tag{16}$$
$$\mu : \mathrm{Pa\,s}. \tag{17}$$

Missing observations are retained as missing values. They are not interpreted as zero.

4 Descriptive Analysis of Thermophysical Properties

4.1 The 20 Degree Celsius Comparison

Table 2 presents the manufacturer-reported properties at 20°C.

Table 2: Dynalene HC Thermophysical Properties at 20°C

Grade	c_p **(kJ/kg K)**	k **(W/m K)**	ρ **(kg/m^3)**	μ **(mPa s)**
HC-10	3.320	0.524	1190	1.80
HC-20	3.212	0.523	1237	2.10
HC-30	3.077	0.519	1275	2.50
HC-40	2.930	0.509	1315	2.70
HC-50	2.701	0.505	1340	3.20

The reported values display a consistent descriptive pattern within this five-grade cohort.

Across the observations at 20°C, increasing grade number is associated with:

1. lower specific heat capacity;
2. lower thermal conductivity;
3. higher density;
4. higher dynamic viscosity.

These are descriptive relationships in the manufacturer-reported tables. They do not establish a universal chemical law or a causal relationship between an undisclosed concentration fraction and thermophysical performance.

HC-10 has the highest reported specific heat capacity and thermal conductivity in this comparison. It also has the lowest reported density and viscosity.

HC-50 has the lowest reported specific heat capacity and thermal conductivity, together with the highest reported density and viscosity.

These differences motivate the subsequent engineering analysis.

4.2 Heat-Capacity Rate

For a single-phase fluid under approximately constant specific heat capacity, the heat-capacity rate is:

$$\dot{C}_p = \dot{m} c_p, \tag{18}$$

where $\dot{m}$ denotes mass flow.

For a specified mass-flow rate, a greater specific heat capacity produces a greater heat-capacity rate.

This relationship is a thermodynamic identity under the stated approximation. It does not independently establish the total heat-transfer rate of an exchanger.

4.3 The Importance of Temperature Dependence

A comparison at one temperature is insufficient for evaluating a coolant intended to operate over a range of thermal conditions.

The relevant property functions are:

$$\mathbf{q}_g(T) = \begin{pmatrix} c_{p,g}(T) \\ k_g(T) \\ \rho_g(T) \\ \mu_g(T) \end{pmatrix}. \tag{19}$$

The operating performance of a coolant depends on the joint evolution of these properties.

For example, a change in viscosity affects Reynolds number and pressure loss, while a change in specific heat capacity affects the heat-capacity rate.

Consequently, the temperature dependence of the property vector must be considered jointly rather than through isolated comparisons.

5 Thermohydraulic Model

5.1 Operating Domain

The performance-surface model uses five coolant grades, 22 common exact temperatures, and three mass-flow setpoints.

The common temperatures are:

$$T \in \{-10, 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, \tag{20}$$

$$110, 120, 130, 140, 150, 160, 170, 180, 190, 200\}. \tag{21}$$

The mass-flow setpoints are:

$$\dot{m} \in \{0.2, 0.3, 0.4\} \text{ kg/s}. \tag{22}$$

The model assumes:

$$D = 0.020 \text{ m}, \tag{23}$$
$$L = 5.0 \text{ m}, \tag{24}$$
$$\epsilon = 1.5 \times 10^{-6} \text{ m}, \tag{25}$$
$$\eta_p = 0.65, \tag{26}$$

where D is internal pipe diameter, L is pipe length, ϵ is absolute roughness, and η_p is assumed pump efficiency.

The total number of operating points is:

$$N = 5 \times 22 \times 3 = 330. \tag{27}$$

The common-temperature restriction ensures that each grade is compared using an exact manufacturer-reported property vector at the same temperature.

No property interpolation or extrapolation is performed in constructing this surface.

5.2 Continuity and Velocity

For a circular pipe, the cross-sectional area is:

$$A = \frac{\pi D^2}{4}. \tag{28}$$

The mean fluid velocity is:

$$v = \frac{\dot{m}}{\rho A}. \tag{29}$$

5.3 Reynolds Number

The Reynolds number is:

$$Re = \frac{\rho v D}{\mu}. \tag{30}$$

Substituting the expression for velocity gives:

$$Re = \frac{4\dot{m}}{\pi D \mu}. \tag{31}$$

For the specified geometry and mass-flow constraint, Reynolds number depends directly on mass flow and inversely on dynamic viscosity.

5.4 Prandtl Number

The Prandtl number is:

$$Pr = \frac{\mu c_p}{k}, \tag{32}$$

where specific heat capacity must be expressed in consistent SI units.

The Prandtl number characterizes the relative diffusion of momentum and thermal energy.

5.5 Darcy Friction Factor

The Darcy friction factor is estimated using the Churchill all-regime expression.

Define:

$$A_C = \left[2.457 \ln\left(\frac{1}{(7/Re)^{0.9} + 0.27(\epsilon/D)}\right)\right]^{16}, \tag{33}$$

$$B_C = \left(\frac{37530}{Re}\right)^{16}. \tag{34}$$

Then:

$$f_D = 8\left[\left(\frac{8}{Re}\right)^{12} + \frac{1}{(A_C + B_C)^{3/2}}\right]^{1/12}. \tag{35}$$

The model uses the Darcy friction factor, not the Fanning friction factor.

5.6 Pressure Loss

The pressure loss is calculated using the Darcy–Weisbach equation:

$$\Delta P = f_D \frac{L}{D}\frac{\rho v^2}{2}. \tag{36}$$

This calculation represents straight-pipe frictional loss under the assumed geometry.

It does not include fittings, manifolds, valves, filters, heat exchangers, cold plates, connectors, or elevation changes.

5.7 Pumping Power

The volumetric flow rate is:

$$\dot{V} = \frac{\dot{m}}{\rho}. \tag{37}$$

The hydraulic power is:

$$P_h = \Delta P \dot{V}. \tag{38}$$

The estimated electrical pump input is:

$$P_e = \frac{\Delta P \dot{V}}{\eta_p}. \tag{39}$$

The pump efficiency is fixed at 65 percent for the screening exercise.

This is an illustrative assumption, not a measured pump characteristic.

5.8 Heat-Capacity Rate

The heat-capacity rate is:

$$\dot{C}_p = \dot{m} c_p. \tag{40}$$

For reporting in watts per kelvin, specific heat capacity is converted from kilojoules per kilogram kelvin to joules per kilogram kelvin.

5.9 Convective Heat Transfer

The internal convective heat-transfer coefficient is estimated using the Gnielinski correlation.

The Nusselt number is:

$$Nu = \frac{(f_D/8)(Re - 1000)Pr}{1 + 12.7\sqrt{f_D/8}\left(Pr^{2/3} - 1\right)}. \tag{41}$$

The internal heat-transfer coefficient is:

$$h_i = \frac{Nu\, k}{D}. \tag{42}$$

The model calculates these quantities only when:

$$3000 \leq Re \leq 5,000,000, \tag{43}$$

$$0.5 \leq Pr \leq 2000. \tag{44}$$

Operating points outside the selected nominal correlation envelope are flagged. They are not silently extrapolated.

6 The Thermohydraulic Performance Surface

6.1 Vector-Valued Performance

The principal engineering output is a vector-valued performance function:

$$\mathbf{P}_g(T, \dot{m}; D, L, \epsilon, \eta_p) = \begin{pmatrix} h_i \\ \Delta P \\ P_e \\ \dot{m}c_p \end{pmatrix}. \tag{45}$$

This representation is preferable to a single scalar performance indicator because the components represent different physical and operational objectives.

The heat-transfer coefficient, pressure loss, pump input, and heat-capacity rate cannot be treated as interchangeable quantities.

6.2 Performance at 20 Degrees Celsius

Table 3 presents the model-estimated results at 20°C and a mass-flow rate of 0.3 kg/s.

Table 3: Model-Estimated Performance at 20°C and 0.3 kg/s

Grade	**Re**	ΔP **(kPa)**	P_e **(W)**	$\dot{C}_p$ **(W/K)**
HC-10	10610	2.93	1.14	996
HC-20	9095	2.94	1.10	964
HC-30	7639	3.00	1.09	923
HC-40	7074	2.97	1.04	879
HC-50	5968	3.06	1.06	810

Note: The corresponding model-estimated internal heat-transfer coefficients are 2621, 2365, 2084, 1926, and 1665 W/m^2K for HC-10 through HC-50, respectively. All reported operating points are within the selected Gnielinski nominal envelope. Values are model outputs, not physical test measurements.

The results demonstrate that different engineering objectives produce different comparisons.

At this operating point, HC-10 has the highest estimated internal heat-transfer coefficient and heat-capacity rate.

HC-40 has the lowest estimated pump electrical input.

HC-50 has the highest estimated pressure loss.

The differences are not sufficiently interpreted by assigning a universal rank to the five grades. Their relative usefulness depends on the system objective and constraints.

6.3 Performance-Surface Ranges

Across the 330 modeled operating points, the reported ranges are:

$$Re \in [2126, 82144], \tag{46}$$

$$\Delta P \in [0.99, 5.97] \text{ kPa}, \tag{47}$$

$$P_e \in [0.26, 2.80] \text{ W}. \tag{48}$$

The internal heat-transfer coefficient is reported only for points within the selected Gnielinski applicability envelope.

Four of the 330 operating points are outside that envelope.

The remaining 326 points are assigned a model-estimated internal heat-transfer coefficient under the selected correlation.

The four excluded points remain in the dataset for the other calculations, but their internal heat-transfer coefficients are not supplied by this model.

6.4 Interpretation of the Surface

The performance surface establishes three important points.

First, temperature affects the thermophysical properties that determine the operating response.

Second, mass flow changes both the hydraulic and thermal operating conditions.

Third, the comparative performance of coolant grades is conditional on the selected operating point.

A ranking at 20°C and 0.3 kg/s cannot automatically be generalized to the entire temperature-flow domain.

The performance surface is therefore a conditional engineering map rather than a universal product ranking.

7 Economic Interpretation of Thermohydraulic Performance

7.1 From Physical Performance to Economic Cost

The thermohydraulic model provides physical quantities that can enter an economic evaluation.

The estimated pump input is one component of operating energy consumption.

The heat-capacity rate and convective heat-transfer coefficient help characterize the ability of a system to deliver a specified thermal service.

However, neither quantity directly measures the economic value of the service.

The economic comparison requires a common service definition.

For example, two coolant-system configurations should be compared under a specified heat load, inlet condition, outlet-temperature requirement, and reliability constraint.

Comparing pump input alone would be insufficient if the configurations deliver different cooling services.

7.2 Lifecycle Cost

For coolant-system configuration g, define lifecycle cost as:

$$LCC_g = C_{0,g} + \sum_{t=1}^{H} \frac{C_{E,g,t} + C_{M,g,t} + C_{R,g,t} + C_{F,g,t} + C_{D,g,t}}{(1+r)^t}, \tag{49}$$

where:

$$C_{0,g} : \text{initial capital and installation cost,} \tag{50}$$
$$C_{E,g,t} : \text{energy cost,} \tag{51}$$
$$C_{M,g,t} : \text{maintenance cost,} \tag{52}$$
$$C_{R,g,t} : \text{fluid replacement cost,} \tag{53}$$
$$C_{F,g,t} : \text{failure and downtime cost,} \tag{54}$$
$$C_{D,g,t} : \text{disposal and compliance cost,} \tag{55}$$
$$r : \text{discount rate,} \tag{56}$$
$$H : \text{evaluation horizon.} \tag{57}$$

The cost terms must be measured or estimated for a clearly specified application.

The current dataset does not contain the required commercial prices, capital costs, maintenance histories, electricity tariffs, or failure probabilities. Consequently, no numerical lifecycle-cost ranking is reported.

7.3 Levelised Cost of Cooling

A service-normalized cost measure can be defined as:

$$LCOC_g = \frac{LCC_g}{\sum_{t=1}^{H} \frac{Q_{g,t}}{(1+r)^t}}, \tag{58}$$

where $Q_{g,t}$ is the useful cooling service delivered during period t.

The units of $Q_{g,t}$ must be defined consistently for the application.

For example, an industrial process may measure useful thermal duty in kilowatt-hours of heat removed, while a data-centre application may additionally require a specified inlet-temperature range, equipment availability, and service continuity.

The levelised cost of cooling is meaningful only when the compared configurations deliver equivalent services under a common system boundary.

7.4 The Economic Decision Rule

The economically preferred configuration is not necessarily the one with the highest heat-transfer coefficient.

A general decision rule is:

$$g^* = \arg\min_{g \in \mathcal{G}} LCC_g \tag{59}$$

subject to:

$$Q_{g,t} \geq Q_{\min,t}, \tag{60}$$
$$T_{\text{out},g,t} \leq T_{\max,t}, \tag{61}$$
$$P_{e,g,t} \leq P_{\max,t}, \tag{62}$$
$$R_{g,t} \geq R_{\min,t}, \tag{63}$$

where the constraints represent minimum useful cooling, maximum permissible outlet temperature, an auxiliary-power limit, and a reliability requirement.

Additional constraints may be imposed for corrosion, toxicity, flammability, electrical compatibility, environmental compliance, and material compatibility.

The present performance surface does not evaluate these additional dimensions.

8 The Cooling Dividend Function: Empirical Connection

8.1 Gross Cooling Dividend

The Cooling Dividend Function introduced in the preceding theoretical treatise is:

$$D_C = \mathcal{D}(C, \Theta, \mathbf{X}, \mathcal{A}), \tag{64}$$

where:

$$C : \text{effective cooling service}, \tag{65}$$
$$\Theta : \text{initial thermal exposure}, \tag{66}$$
$$\mathbf{X} : \text{economic and infrastructural characteristics}, \tag{67}$$
$$\mathcal{A} : \text{cooling architecture}. \tag{68}$$

The present study investigates the engineering characteristics of commercial fluids that may influence C.

It does not directly observe the economic output difference required to estimate D_C.

8.2 Marginal Cooling Dividend

The marginal Cooling Dividend is:

$$MD_C(C) = \frac{\partial D_C(C)}{\partial C}. \tag{69}$$

The engineering performance surface may inform the technological relationship between coolant choice and delivered cooling, but it does not independently identify the marginal economic value of an additional unit of cooling.

That value requires economic outcome data and a credible counterfactual.

8.3 Net Cooling Dividend

The Net Cooling Dividend is:

$$N_C(C) = D_C(C) - SC(C). \tag{70}$$

The social opportunity cost may include energy, water, materials, capital, environmental externalities, and other resources required to provide cooling.

The economic analysis must avoid double-counting costs already reflected in market prices or in the valuation of other components.

8.4 A Coolant-Specific Representation

For an application with a specified cooling-service requirement, the economic contribution of a coolant-system configuration can be represented as:

$$D_{C,g} = \mathcal{D}\left(C_g, \Theta, \mathbf{X}, \mathcal{A}_g\right). \tag{71}$$

The subscript g identifies the coolant-system configuration.

This expression does not imply that the fluid alone determines the Cooling Dividend. The architecture, equipment, operating conditions, and delivered service remain part of the system.

A more complete empirical design would compare alternative configurations delivering the same cooling service and estimate their differences in lifecycle cost, reliability, and economic output.

9 Empirical Hypotheses

The empirical and computational framework motivates the following hypotheses.

9.1 Hypothesis 1: Temperature-Dependent Thermophysical Performance

The thermophysical properties of a commercial coolant vary with temperature, producing temperature-dependent changes in its thermohydraulic response.

This hypothesis is investigated descriptively through the manufacturer-reported property observations.

9.2 Hypothesis 2: Grade Heterogeneity

The five Dynalene HC grades exhibit different thermophysical and modeled thermohydraulic performance profiles.

The 20-degree comparison and the 330-point performance surface provide descriptive and model-based evidence relevant to this hypothesis.

9.3 Hypothesis 3: Hydraulic-Thermal Trade-Off

A coolant with a higher estimated internal heat-transfer coefficient does not necessarily minimize pressure loss or pump input.

The 20°C, 0.3 kg/s operating point illustrates the possibility of different outcomes across performance dimensions.

9.4 Hypothesis 4: Operating-Condition Dependence

The relative performance of coolant grades depends on temperature, mass flow, geometry, and the applicable heat-transfer model.

The common-temperature performance surface provides a framework for investigating this conditional dependence.

9.5 Hypothesis 5: Economic Objective Dependence

The coolant configuration that maximizes a selected thermal performance measure need not minimize lifecycle cost or maximize net economic cooling value.

This hypothesis requires application-level economic data and remains to be tested empirically.

10 Statistical and Computational Validation

10.1 Property-Function Estimation

The property observations may be used to estimate temperature-dependent functions:

$$c_{p,g}(T) = f_{c,g}(T) + u_{c,g}(T), \tag{72}$$

$$k_g(T) = f_{k,g}(T) + u_{k,g}(T), \tag{73}$$

$$\rho_g(T) = f_{\rho,g}(T) + u_{\rho,g}(T), \tag{74}$$

$$\mu_g(T) = f_{\mu,g}(T) + u_{\mu,g}(T). \tag{75}$$

The functions should be estimated separately for each exact formulation.

Candidate specifications include low-order polynomials for selected properties and nonlinear functional forms for viscosity.

Model selection should consider:

1. residual diagnostics;
2. physical plausibility;
3. out-of-sample prediction;
4. stability across the documented temperature range;
5. the risk of extrapolation.

The current common-temperature performance surface does not require property-function estimation because it uses exact manufacturer-reported observations.

10.2 Validation Metrics

For a property or performance model evaluated against independent observations, standard predictive metrics include:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}. \tag{76}$$

The mean absolute error is:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i|. \tag{77}$$

These metrics require an appropriate validation dataset.

The present model has not been independently validated against a physical test loop. Therefore, no empirical RMSE or MAE is claimed for the thermohydraulic performance surface.

10.3 Avoiding Data Leakage

When property functions are estimated, random splitting of individual temperature observations may produce overly optimistic validation if observations from the same formulation appear in both training and testing samples.

A more appropriate strategy is to hold out temperature regions or complete formulations, depending on the prediction objective.

External validation should use independent measurements rather than additional transcriptions from the same manufacturer document.

10.4 Uncertainty

The manufacturer tables do not provide measurement uncertainty or a harmonized measurement protocol for the present cohort.

The computational model also contains structural uncertainty arising from:

1. assumed pipe geometry;
2. assumed surface roughness;
3. fixed pump efficiency;
4. correlation selection;
5. property measurement uncertainty;
6. omitted system components;
7. constant-property treatment within each operating point.

The numerical outputs should therefore be interpreted as conditional model estimates.

A complete uncertainty analysis should propagate uncertainty in the input properties and engineering assumptions through the performance equations.

11 Limitations

The results are subject to several limitations.

11.1 Manufacturer-Reported Data

The property observations are not independent laboratory measurements.

The HC-10, HC-20, HC-30, and HC-40 observations are sourced from the 2025 engineering guide, while HC-50 is sourced from the 2020 technical data sheet.

The source revisions and measurement protocols are not harmonized.

11.2 Unspecified Measurement Conditions

The source tables do not provide measurement uncertainty or detailed test-method information for the property observations used here.

The pressure basis of the thermophysical-property tables is also not stated.

11.3 Grade Concentration

The grade suffixes do not establish a disclosed mass or volume concentration fraction.

The present analysis therefore treats each grade as an exact commercial identifier without assigning an unsupported concentration basis.

11.4 Simplified Pipe Geometry

The model represents a straight circular pipe.

It excludes fittings, valves, manifolds, heat exchangers, cold plates, filters, connectors, and elevation changes.

The calculated pressure losses are therefore not complete system pressure losses.

11.5 Fixed Pump Efficiency

The pump efficiency is assumed to be 65 percent.

Actual pump performance depends on the pump curve, operating point, motor, control system, and system curve.

A realistic economic comparison requires coupling the hydraulic model to actual equipment characteristics.

11.6 Heat-Transfer Correlation

The Gnielinski correlation is used only within the selected nominal applicability envelope.

The model does not establish whole-exchanger performance, thermal stability, or a guaranteed heat-transfer duty.

11.7 Constant Properties Within Operating Points

The model uses the property vector reported at each exact temperature.

It does not solve the complete coupled thermal trajectory along the pipe or update properties continuously as the fluid changes temperature.

11.8 Absence of Economic Outcome Data

The current dataset does not contain:

1. observed economic output attributable to cooling;
2. measured cooling-service delivery at an application level;
3. electricity tariffs or energy-price histories;
4. coolant purchase and replacement prices;
5. capital equipment costs;
6. maintenance and failure histories;
7. downtime valuations;
8. environmental compliance and disposal costs.

Consequently, lifecycle costs, levelised cooling costs, and monetary Cooling Dividends cannot yet be estimated from the current dataset alone.

11.9 No Safety or Compatibility Certification

The present analysis does not establish corrosion resistance, toxicity, flammability, electrical compatibility, environmental persistence, or material compatibility.

These characteristics require separate source evidence and, where appropriate, controlled testing.

12 Proposed Experimental Validation

A controlled test programme is necessary to determine whether the modeled performance differences are reproduced under physical operating conditions.

12.1 Test Configuration

A suitable experimental apparatus should include:

1. a calibrated circulation loop;
2. a controlled heat-input source;
3. calibrated inlet and outlet temperature sensors;
4. a mass-flow measurement instrument;
5. differential-pressure measurement;
6. electrical power measurement for the pump;
7. documented pipe and heat-exchanger geometry;
8. controlled coolant composition and sample identification.

12.2 Operating Conditions

The five grades should be evaluated under at least three distinct operating constraints:

1. equal mass flow;
2. equal volumetric flow;
3. equal pump electrical power.

These comparisons answer different engineering questions and should not be merged into one undifferentiated ranking.

12.3 Repeated Measurements

Each operating condition should be repeated sufficiently to estimate measurement variability.

At least three repetitions per condition provide an initial minimum for assessing repeatability, although a formal sample-size calculation should determine the final experimental design.

12.4 Uncertainty Propagation

The experimental programme should report instrument accuracy, calibration records, repeatability, and propagated uncertainty for:

$$\dot{m}, \quad T_{\mathrm{in}}, \quad T_{\mathrm{out}}, \quad \Delta P, \quad P_e, \quad \dot{Q}. \tag{78}$$

The measured thermal duty for a single-phase fluid may be estimated using:

$$\dot{Q} = \dot{m}(h_{\mathrm{out}} - h_{\mathrm{in}}). \tag{79}$$

Under approximately constant specific heat capacity:

$$\dot{Q} \approx \dot{m} c_p (T_{\mathrm{out}} - T_{\mathrm{in}}). \tag{80}$$

The enthalpy formulation should be preferred when property variation across the temperature interval is material.

13 An Expanded Economic Research Architecture

The thermohydraulic analysis is one component of a larger empirical programme.

The complete economic architecture can be represented as:

$$\text{Thermal Exposure} \longrightarrow \text{Cooling Demand} \longrightarrow \text{Cooling Architecture} \longrightarrow \text{Delivered Cooling} \longrightarrow \text{Economic Outcomes}. \tag{81}$$

Each stage requires separate variables and evidence.

13.1 Cooling Demand

Cooling demand depends on environmental thermal exposure, production technology, building characteristics, occupancy, equipment density, and required operating temperatures.

13.2 Cooling Architecture

Cooling architecture determines how thermal services are produced and distributed.

Its characteristics include equipment configuration, working fluid, flow network, heat exchangers, control systems, and redundancy.

13.3 Delivered Cooling

Delivered cooling should be measured through useful thermal duty and service quality rather than nominal equipment capacity alone.

Relevant variables include thermal duty, temperature stability, service availability, energy consumption, and system reliability.

13.4 Economic Outcomes

The economic consequences may include:

1. industrial production and productivity;
2. equipment preservation and asset life;
3. labour productivity;
4. household welfare;
5. digital infrastructure reliability;
6. energy and water resource allocation;
7. maintenance and replacement expenditure.

These outcomes should not be mechanically added together. Intermediate transactions, indirect effects, and welfare measures require consistent accounting to avoid double-counting.

13.5 Future Megacity Panel

A complementary empirical programme can examine urban thermal exposure across a defined megacity sample.

Historical weather data could be combined with urban population, cooling access, electricity consumption, infrastructure, and economic indicators.

The relevant empirical unit would be an urban area observed over time.

A possible panel specification is:

$$Y_{it} = \alpha_i + \lambda_t + \beta C_{it} + \gamma \Theta_{it} + \delta(C_{it}\Theta_{it}) + \mathbf{X}'_{it}\boldsymbol{\eta} + \varepsilon_{it}, \tag{82}$$

where:

$$Y_{it} : \text{economic outcome in city } i \text{ and period } t, \tag{83}$$
$$C_{it} : \text{effective delivered cooling service}, \tag{84}$$
$$\Theta_{it} : \text{thermal exposure}, \tag{85}$$
$$\alpha_i : \text{city fixed effects}, \tag{86}$$
$$\lambda_t : \text{time fixed effects}, \tag{87}$$
$$\mathbf{X}_{it} : \text{observed control variables}. \tag{88}$$

The interaction term allows the relationship between cooling and economic outcomes to vary with thermal exposure.

This specification is a proposed research design, not an estimated result of the present treatise.

Causal interpretation would require a defensible identification strategy addressing endogenous cooling investment, income, urban development, energy access, and adaptation.

14 General Findings and Propositions

The empirical and computational investigation supports the following findings and conditional propositions.

14.1 Finding 1: Thermophysical Properties Are Multidimensional

The five-grade comparison demonstrates that specific heat capacity, thermal conductivity, density, and viscosity differ across the selected commercial grades.

No single property provides a complete measure of cooling-system performance.

14.2 Finding 2: Performance Is Operating-Condition Dependent

The thermohydraulic response depends on temperature, mass flow, geometry, and the relevant engineering correlations.

A result obtained at one operating point cannot be generalized automatically to the entire operating domain.

14.3 Proposition 1: Conditional Thermohydraulic Ranking

For two coolant configurations g and j, a ranking based on one performance component need not imply the same ranking for another component.

Formally, it is possible that:

$$h_g > h_j, \tag{89}$$
$$P_{e,g} > P_{e,j}. \tag{90}$$

The proposition follows from the multidimensional structure of the thermohydraulic response. It does not require that this inequality hold for every pair of fluids or operating conditions.

14.4 Proposition 2: The Economic Ranking Is Objective-Dependent

If alternative coolant configurations differ in thermal performance, hydraulic cost, capital requirements, reliability, or lifecycle expenditure, their economic ranking depends on the service requirement and the valuation of these components.

Therefore, a universal ranking based exclusively on a thermophysical property is not generally sufficient for economic selection.

14.5 Proposition 3: Thermohydraulic Equivalence Is Conditional

Two coolant-system configurations may be considered thermohydraulically equivalent only with respect to a specified operating domain, response vector, and tolerance.

Let:

$$\mathbf{y}_g = \begin{pmatrix} \dot{Q}_g \\ \Delta P_g \\ P_{e,g} \\ T_{\max,g} \end{pmatrix}. \tag{91}$$

A conditional equivalence criterion may be written:

$$d(\mathbf{y}_g, \mathbf{y}_j) \leq \tau, \tag{92}$$

where d is a declared distance measure and τ is a pre-specified tolerance.

The criterion does not establish equivalence in chemistry, safety, compatibility, environmental impact, or lifecycle cost.

14.6 Proposition 4: Economic Cooling Requires Service-Normalized Evaluation

A comparison of cooling configurations is economically meaningful only when the delivered service and relevant system boundary are specified.

A lower pump input is not sufficient to establish lower cost per unit of useful cooling if the configurations deliver different thermal duties.

14.7 Proposition 5: The Cooling Dividend Is Not a Thermophysical Property

The Cooling Dividend is an economic quantity.

Thermophysical properties and thermohydraulic outputs may influence its realization through the delivered cooling service, but they are not themselves measurements of the dividend.

A monetary Cooling Dividend requires economic outcomes and an appropriate counterfactual.

15 Conclusion

This treatise has developed an empirical and computational framework for investigating the thermohydraulic and economic performance of commercial coolants.

The analysis uses a source-audited property dataset containing 137 manufacturer-reported observations and constructs a comparative performance surface with 330 model-estimated operating points across five Dynalene HC grades, 22 common exact temperatures, and three mass-flow setpoints.

The results demonstrate that commercial coolant performance is multidimensional and operating-condition dependent.

At the illustrative operating point of 20°C and 0.3 kg/s, HC-10 has the highest estimated internal heat-transfer coefficient and heat-capacity rate among the five grades, while HC-40 has the lowest estimated pump electrical input. These differences demonstrate why the selection of a coolant cannot be reduced to a single thermophysical ranking.

The model also illustrates the importance of correlation validity. Four of the 330 operating points fall outside the selected Gnielinski nominal applicability envelope, and their internal heat-transfer coefficients are appropriately left unreported.

The economic analysis establishes a framework for lifecycle cost and levelised cost of cooling, but the current data do not support numerical estimates of these quantities. More importantly, they do not yet permit direct estimation of the monetary Cooling Dividend.

The next empirical stage must therefore combine thermohydraulic modelling with controlled physical measurements and application-level economic data.

The larger research objective is to connect thermal exposure, cooling demand, cooling architecture, delivered cooling, and economic outcomes within one coherent empirical framework.

The central conclusion is:

$$\boxed{\text{The economic value of cooling depends on the value of the service delivered,}} \tag{93}$$

$$\boxed{\text{relative to the full resource and opportunity cost of delivering it.}} \tag{94}$$

Cooling should not be maximized without regard to its resource requirements. It should be evaluated and optimized as an economic service.

Data Availability and Reproducibility

The source data, product catalogue, source register, computational scripts, and generated performance-surface dataset are maintained in the EconomicCooling repository:

https://github.com/TheRealoliGARCH/EconomicCooling

The principal files used in this study are:

`data/coolant_observations.csv` Master manufacturer-reported property observations.

`data/products.csv` Product and formulation catalogue.

`data/source_register.csv` Source provenance register.

`reports/hc_performance_surface.csv` The 330 model-estimated operating points.

`scripts/build_hc_performance_surface.py` Reproducible performance-surface construction script.

`docs/DATA_DICTIONARY.md` Data definitions, units, and evidence classifications.

`docs/METHODOLOGY.md` Modelling and validation methodology.

`reports/DYNALENE_HC_PERFORMANCE_SURFACE.md` Performance-surface assumptions, results, and limitations.

The performance-surface script uses the Python standard library. It selects exact temperatures common to all five grades and does not interpolate or extrapolate thermophysical properties.

The generated performance surface is a model output. It must not be interpreted as a set of independent experimental measurements.

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Glossary

Cooling Architecture:
The configuration of equipment, fluids, networks, controls, and operating arrangements used to deliver cooling services.

Cooling Dividend:
The incremental economic value attributable to a specified cooling service relative to a defined counterfactual.

Delivered Cooling:
The useful thermal service actually provided by a cooling system.

Gross Cooling Dividend:
The incremental economic value before deducting the incremental social opportunity cost of cooling.

Heat-Capacity Rate:
The product of mass-flow rate and specific heat capacity.

Lifecycle Cost:
The discounted cost of acquiring, operating, maintaining, replacing, and disposing of a cooling-system configuration over a specified horizon.

Marginal Cooling Dividend:
The additional economic value attributable to an incremental unit of cooling service.

Net Cooling Dividend:
The Gross Cooling Dividend less the incremental social opportunity cost of providing cooling.

Thermohydraulic Performance:
The combined thermal and hydraulic response of a fluid-system configuration under specified operating conditions.

Thermohydraulic Equivalence:
Conditional similarity between system responses over a declared operating domain and within specified tolerances.

The End